

Objective Data

1.1 Dihexa 10pM or 100pM reduced the impairment of spatial learning by Scopolamine. High dose (100pM) Dihexa was able to completely overcome the scopolamine-induced deficits. Oral Dihexa at 1.25mg/kg/day or 2mg/kg/day for 8d had similar effects on preventing the impairment by Scopolamine, with the 2mg/kg/day completely abolishing the Scopolamine-induced deficits.

1.2 In aged rats, Dihexa 2mg/kg/day PO enhanced cognitive performance relative to control.

1.3 Dihexa dose-dependently stimulates spinogenesis. Dihexa increases spine density in beta-Actin transfected hippocampal neurons dose-dependently. 1pM of Dihexa stimulates spinogenesis.

1.4 Dihexa increases spine head width, as well as correlation between presynaptic and postsynaptic connections. This suggests functional synapses are formed.

1.5 1pM of Dihexa for 5d increased the frequency of miniature excitatory postsynaptic currents (mEPSCs), mediated by AMPAR, and which were not blocked by Strychnine, Picrotoxin, or Tetrodotoxin (TTX).

1.6 Dihexa causes spinogenesis in CA1 hippocampal neurons. 1pM Dihexa is similar in magnitude of effect to 10ng/mL (119pM) hepatocyte growth factor (HGF).

1.7 100fM Dihexa is insufficient to produce spinogenesis.

1.8 Hinge, an HGF inhibitor, reduced the effect of HGF and Dihexa. Dihexa stimulated c-Met phosphorylation.

2.1 50uM BDNF produced similar effects to 1pM of Nle¹-AngIV.

2.2 Angiotensin IV (AngIV) is Val¹-Tyr²-Ile³-His⁴-Pro⁵-Phe⁶. Thus, Dihexa is a modified AngIV²⁻³ analogue.

2.3 AngIV increased spine numbers and enlarged spine head size.

2.4 AngIV and multiple AngIV analogues facilitate LTP, learning, and memory consolidation, increase cerebral blood flow (CBF), and provide neuroprotection. Nle¹-AngIV reverses deficit in dementia models induced by: treatment with Scopolamine, Kainate injections into the hippocampus, perforant path cuts, and ischemia resulting from transient 4-vessel occlusion. Brain binding sites for [¹²⁵I]AngIV have been autoradiographically localized in structures known to mediate cognitive processing including the neocortex, hippocampus, and basal nucleus of Meynert.

2.5 One explanation for the pro-cognitive activity of AngIV-related molecules is a capacity to expand synaptic connectivity and augment synaptic communication.

2.6 BDNF was used as a positive control. BDNF, widely recognized as a strong potentiator of dendritic spine plasticity and synaptic strength in hippocampal neurons, induced alternations in spine architecture equivalent to those observed with pro-cognitive AngIV and related peptides. However, it should be noted that the concentration of BDNF required to produce equivalent increases in dendritic spines was 7 orders of magnitude greater than that needed for Nle¹-AngIV.

3.1 A phosphorylated Dihexa prodrug, Fosgonimeton, causes dose-dependent increases in Gamma power in healthy humans, and improves P300 latency. Fosgonimeton decreases NfI and GFAP.

3.2 AChEIs may inhibit the beneficial effects of Fosgonimeton.

4.1 Unlike native HGF, Dihexa is chemically stable and blood-brain barrier (BBB)-penetrant. As a synthetic HGF mimetic, it forms a functional ligand by dimerizing with endogenous HGF to activate the HGF receptor and its downstream signaling cascades.

4.2 Dihexa protected larval zebrafish lateral hair cells from 2 different ototoxins, Neomycin and Gentamicin. A Dihexa concentration of 1uM confers optimal protection from acute treatment with either ototoxin.

4.3 Dihexa-mediated protection is attenuated by co-treatment with the HGF antagonist 6-AH, which is evidence that HGF activation is a component of the observed protection. Dihexa's robust protection is partially attenuated with inhibitors of the downstream HGF targets Akt, mTOR, and MEK.

4.4 Addition of an amino group to the N-terminal of Dihexa also attenuates the protective response, suggesting that even small substitutions greatly alter the specificity of Dihexa for its target.

4.5 HGF is a potent neurotrophic factor that promotes a variety of neuronal cell types to promote axon guidance and cell survival.

4.6 Dihexa is capable of stimulating HGF and its associated receptor c-Met.

4.7 Dihexa is a synthetically derived AngIV analog that is BBB-permeable, stable, and orally bioavailable.

4.8 Dihexa ameliorated memory loss in a rat model of Parkinson's Disease (PD).

4.9 Dihexa can bind to and dimerize HGF, forming a functional ligand and thereby activating c-Met. HGF signaling is linked to neuronal protection in rodents both in vitro and in vivo.

4.10 Dihexa protects lateral hair cells from aminoglycoside exposure. It did not alter the entry of aminoglycosides into hair cells but rather attenuates cell death via an HGF-dependent signaling mechanism. The greatest protection was found at 1uM, where only a minor decrease from control hair cell survival scores was observed. An additional peak of protection was seen at 100fM (0.1pM), but was inconsistent across experiments. Dihexa-mediated neuroprotection was still robust 6h post-Neomycin washout.

4.11 Activation of the receptor tyrosine kinase (RTK) c-Met leads to the recruitment of many signaling mediators and ultimately the activation of downstream signals, most notably Akt/mTOR and Ras/ERK. The MAPK inhibitor, UO126, and Akt Inhibitor VIII, were used to determine to what extent, if any, activation of their targets is required for Dihexa-mediated protection. Optimal concentrations of UO126 and Akt Inhibitor VIII (1uM) were determined based on the highest concentration of each compound that did not shift the Neomycin dose-response curve. At Neomycin concentrations of 100, 200, and 400uM either inhibitor demonstrated partial attenuation of Dihexa otoprotection, indicating at least partial reliance on their associated signaling proteins for protective effects. When both inhibitors are co-administered the result is similar to each demonstrated independently, suggesting that other signaling proteins may also contribute to protection. Phosphorylation of Akt leads to activation of mTOR, a molecular sensor of metabolism and cellular homeostasis. The mTOR inhibitor Rapamycin to determine the extent to which activation of mTOR is required in Dihexa-mediated protection. 10uM Rapamycin alone did not affect the Neomycin dose-response curve. However, co-treatment with 1uM Dihexa and 10uM Rapamycin showed significant attenuation of hair cell protection at 50 and 100uM Neomycin. Thus, Dihexa-mediated protection relies, at least in part, on the activation of multiple downstream targets of the HGF/c-Met system (Akt/mTOR and Ras/ERK).

4.12 Dihexa failed to protect against the ototoxic effect of Cisplatin in vitro.

4.13 HGF/c-Met activates a diverse set of signal transduction capable of inhibiting cell death. In various culture cell lines, c-Met phosphorylation can result in downstream activation of Akt, ERK/MAPK, JNK, or FAK signaling. Dihexa-mediated hair cell protection requires activation of Akt, mTOR, and MEK signaling. Both HGF application and Akt activity result in increased Bcl-2 expression in cell culture. In human endothelial cells, HGF-induced cellular protection from hypoxia is dependent on Bcl-2 activity.

5.1 Activation of c-Met stimulates mitogenesis, motogenesis, morphogenesis, and the ability to mediate stem cell differentiation and neurogenesis, and protects against tissue insults in a wide range of cells, including neurons. The HGF/c-Met growth factor system has recently been shown to induce dendritic arborization and synaptogenesis when stimulated by Dihexa.

5.2 Dihexa, which is an Angiotensin-based analogue, has the sequence N-Hexanoic-Tyr-Ile-6-aminohexanoic amide. It was derived from the pre-prototype molecule Nle¹-AngIV and has shown promise in facilitating the formation of new functional synaptic connections and augmenting memory consolidation in animal models of Alzheimer's Disease (AD).

5.3 Dihexa facilitates memory consolidation and retrieval.

5.4 HGF, also known as "Scatter Factor" (SF), is a glycoprotein that acts via the Type 1 RTK c-Met. HGF was originally isolated from the liver and shown to promote liver regeneration. The c-Met receptor is made up of disulfide bond-linked alpha (45kDa) and beta (145kDa) subunits. The alpha chain is extracellular while the beta chain is transmembrane. HGF dimerization precedes binding to the c-Met receptor which then undergoes phosphorylation. Once phosphorylated, the tyrosine residues of the beta subunit serve as docking sites for downstream signaling mediators including the extracellular signal-regulated kinase and the PI3K pathway.

5.5 Activation of the HGF/c-Met pathway causes outgrowth and dendritic arborization, facilitated hippocampal LTP and Ca²⁺ signaling, facilitation of cerebral blood flow and cerebroprotection, facilitated memory consolidation, seizure protection, and enhanced wound healing.

5.6 The AngIV/AT₄ system seems to overlap with the HGF/c-Met pathway. Norleual-AngIV's mechanism of action as a c-Met receptor antagonist is by inhibiting the dimerization of HGF, a necessary prerequisite step in order for HGF to bind to and activate the c-Met receptor. This dimerization process is dependent upon a short HGF domain located between its N-terminal and first kringle domain referred to as the hinge region. This hinge region is of critical importance and must be available for dimerization to occur.

5.7 Moderate to heavy brain distribution of the HGF/c-Met (aka AngIV/AT₄) system is found in the amygdala, anterior pituitary, caudate-putamen, cerebellum, globus pallidus, hippocampus, neocortex, nucleus basalis of Meynert, piriform cortex/prefrontal cortex (PFC), substantia nigra (SN)/midbrain, striatum, and thalamus, consistent with effects on cognition. Partial determinations of the brain c-Met receptor distribution generally agree with the above distribution. The AT₄ receptor subtype's ability to facilitate non-NMDAR-dependent LTP suggests a nonglutamatergic signaling pathway involving Ca²⁺ entry. In fact, activation of the AT₄ receptor subtype results in increased internalization of Ca²⁺ via at least 3 different Ca²⁺ channels, resulting in a rapid and salient cell signaling event. HGF-induced responses, including LTP, also depend on internalization of Ca²⁺. Conversion of Angiotensin II (AngII) to AngIV is necessary for AngII-induced dopamine (DA) release in the striatum, and acetylcholine (ACh) release in the hippocampus. Increased neural intracellular Ca²⁺, coupled with Matrix

Metalloproteases (MMPs) released into the extracellular space, suggests a neural plasticity function in agreement with similar observations for HGF. Neural imaging findings indicate that the Nle¹-AngIV stimulates dendritic spine numbers and size in the hippocampus, as well as overall dendritic arborization, suggesting a plausible synaptogenic mechanism to explain the ability of these molecules to enhance synaptic plasticity and connectivity among neurons. HGF, acting at the c-Met receptor, has also been shown to increase dendritic arborization in cultured hippocampal neurons.

5.8 HGF and c-Met are neuronally-expressed in several brain structures including the neocortex and hippocampus, and reveal high densities at excitatory synapses within the hippocampus. Activation of the c-Met receptor by HGF promotes neurite outgrowth and dendritic branching by cortical neurons in slice cultures. HGF increased the size and complexity of dendritic arborization in dissociated hippocampal neurons in culture. This facilitation was blocked by pretreatment with the NMDAR antagonist, DL-2-Amino-5-phosphonopentanoic acid (APV). This HGF effect was dependent upon elevations in intracellular Ca²⁺ and accompanying increases in autophosphorylation of CaMKII. This suggests that Ca²⁺-dependent processes underlies HGF's ability to increase dendritic arborization and are consistent with the observations of increased levels of hippocampal neuronal intracellular Ca²⁺ with Nle¹-AngIV treatment accompanied by facilitated hippocampal dendritic arborization. Pretreatment of cultured hippocampal neurons with an AT₄ receptor antagonist inhibited this Nle¹-AngIV-induced arborization. Activation of the HGF/c-Met system stimulated stem cell proliferation, differentiation, and neurogenesis. This mechanism may be responsible for the protection of Purkinje neurons seen in the degenerative knock-in (KI) mouse model of spinocerebellar ataxia and facilitated endogenous neuron recovery and survival following spinal cord injury (SCI).

5.9 It is possible that the brain HGF/c-Met and insulin-responsive aminopeptidase (IRAP) systems are linked. In support of this hypothesis, the HGF regulates insulin sensitivity and glucose uptake. This appears to occur via the heteromultimerization of the insulin and c-Met receptors to form a functional complex.

5.10 It is likely that Dihexa increases glutamatergic transmission given such a finding using cultured rat hippocampal neurons.

5.11 Nle¹-AngIV, as well as shorter fragments, augmented hippocampal synaptic connectivity via the formation of new synapses. Functionality of these synapses was inferred from analogue-induced spinogenesis and the colocalization of synaptic markers in newly-formed dendritic spines which were coupled with enhanced mEPSCs.

5.12 Dihexa induced spinogenesis/synaptogenesis at pM concentrations, is orally-active, and is slowly cleared from the blood (plasma stability half-life: 335.5min (5.6h)).

5.13 Dihexa binds with high affinity to HGF and induces c-Met phosphorylation in the presence of subthreshold levels of HGF. Dihexa stimulated hippocampal spinogenesis and synaptogenesis equivalent with HGF. Treatment with the HGF antagonist, Hinge, as well as an shRNA directed at c-Met, significantly inhibited these actions. Dihexa penetrated the BBB in sufficient quantity to facilitate memory consolidation and retrieval in the Scopolamine-induced amnesic rat model of AD as well as in aged rats employing the MWM task of spatial memory.

6.1 Exposure to 3-Nitropropionic acid (3-NP), a model of Huntington's disease (HD), decreased the amount of weight rats gained, impaired spatial learning and memory consolidation, and led

to marked motor dysfunction. Dihexa failed to be protective against deficits induced by 3-NP neurotoxicity.

7.1 A combination of Vitamin C, Dihexa, and Forskolin (VDF) could substitute growth factors to induce hepatic differentiation of human pluripotent stem cells (hPSCs). The obtained hepatoblasts (HBs) could subsequently expand and mature into functional hepatocyte-like cells (HLCs).

8.1 The c-Met/HGF system has long been implicated in embryogenesis and organogenesis, and mediates many development processes concerning epithelial-mesenchymal interactions. Thus, during development, HGF is produced by mesenchymal cells in many organs; while c-Met is expressed in nearby epithelial cells. The mesenchymal and epithelial cells exchange signals that result in stimulated epithelial growth, morphogenesis, and differentiation. In adults, HGF/c-Met expression is upregulated in a number of organs including liver, heart, kidney, and promotes cell proliferation and migration and inhibits cellular death. HGF is also involved in liver regeneration. Following hepatic injury, blood and liver HGF levels are significantly elevated. This is true of patients with a number of organ injuries including kidney, lung, and other organs. HGF has also been shown to stimulate mitogenesis and insulin production in pancreatic beta-cells. In fact, beta-cells synthesize HGF which in turn regulates insulin production by beta-cells in order to prevent hyperglycemia. Specific damage to beta-cell c-Met receptors caused reductions in islet size accompanied by drops in blood insulin levels and the onset of mild hyperglycemia in mice. These findings support an important role for the HGF/c-Met system in the regulation of blood glucose levels. HGF is also required for repairing injuries of skin, muscle, and cartilage. With each injury, HGF is initially secreted as a pro-HGF molecule and then converted to its active form at the injury site by "HGF-activators". This provides a mechanism that permits c-Met receptor activation at the site of organ injury.

8.2 HGF and c-Met are overexpressed in a variety of solid tumors and can result in facilitated tumor growth and malignancy. As a result, HGF/c-Met presence often correlates with poor clinical prognosis. Thus, with multiple mechanisms of activation (proliferation, invasion, angiogenesis, and anti-apoptosis), the HGF/c-Met system is a major contributor to human cancers. Of related importance, the downregulation of HGF and/or c-Met in human tumor xenografts has been shown to inhibit tumor growth.

8.3 The HGF/c-Met receptor system normally behaves as a tissue self-defense and repair mechanism; however, the increase in HGF production is transient and often delayed resulting in the likelihood of organ failures. Thus, supplemental treatment with HGF may be helpful in alleviating organ disease. Diseases being appreciated in this way include: hepatitis, pneumonia, myocardial infarction (MI), liver cirrhosis, kidney disease, chronic obstructive pulmonary disease (COPD), various inflammatory and immunogenic diseases, and autoimmune diseases.

8.4 In PD, HGF gene transfer has been shown to protect DAergic neurons from a 6-Hydroxydopamine (6-OHDA) challenge.

8.5 In AD, it has been reported that HGF gene treatment reduced amyloid beta (Abeta)-induced cognitive impairment in mice.

8.6 In amyotrophic lateral sclerosis (ALS), HGF gene treatment has been shown to delay disease progression in mouse and rat models.

8.7 There is a small but growing body of literature on the role of HGF in facilitating LTP and memory acquisition and retention. The HGF/c-Met system is a critical factor in the development

of the hippocampus. Inactivation of c-Met in the embryonic proliferative zone of mice causes an increase in Parvalbumin (PV)+ cells in the dentate gyrus, an accompanying loss of CA3 field neurons, with an overall loss of Calretinin+ cells throughout the hippocampus. Thus, HGF/c-Met is required for normal hippocampal development. HGF facilitated LTP in hippocampal slices in a Ca^{2+} -dependent manner.

8.8 Transgenic (Tg) mice overexpressing HGF (HGF-Tg) in a neuron-specific manner displayed superior performance in the Y-maze and MWM tasks of spatial memory compared with age-matched WT controls. The HGF-Tg mice had shorter latencies to find the hidden platform than controls. Further, these mice displayed greater accuracy and persistence to locate the hidden platform than controls.

8.9 The overexpression of HGF resulted in elevated hippocampal levels of the NMDAR subunits NR2A and NR2B (but not NR1) in HGF-Tg mice compared with WT controls. This HGF facilitation of learning and memory performance could potentially be explained by the upregulation of NR2A and NR2B receptors, but there may be further mechanisms.

8.10 HGF regulates proliferation and migration of DAergic progenitor cells isolated from the fetal striatum. These cells were capable of differentiating into functional neurons with the ability to synthesize and release DA. Human embryonic stem cell (hESC)-derived DAergic neurons increased the expression of Tyrosine Hydroxylase (TH; a DA neuron marker) with the application of any one of several growth factors to the cell culture including HGF, stromal cell-derived factor-1 α , and VEGF. Thus, growth factors may be of potential use to induce DA cellular differentiation of hPSCs.

8.11 HGF is a potent neurotrophic factor for many neuronal cell types, including motoneurons, hippocampal neurons, cerebellar granular cells, and sympathetic neurons.

8.12 HGF has been shown to positively impact ischemic-induced injuries such as cardiac and hind limb ischemia. HGF has also been shown to eliminate hippocampal neuronal cell loss in transient global cerebral ischemic gerbils and transient focal ischemic rats. HGF-induced improvement in escape latencies by microsphere embolism-cerebral ischemic rats using a circular water maze task. There was reduced damage to cerebral endothelial cells in ischemic animals treated with HGF. Overexpression of HGF following permanent middle cerebral artery occlusion (pMCAO) resulted in significant recovery of performance in the MWM and passive avoidance conditioning tasks. HGF was found to increase the number of arteries in the neocortex some 50d following the onset of ischemia.

8.13 HGF and c-Met are neuronally-expressed including the neocortex and hippocampus, and appear at high densities at excitatory synapses within the hippocampus. Activation of c-Met by HGF promotes neurite outgrowth and dendritic branching by cortical neurons in slice cultures. HGF increased the size and complexity of dendritic arborization in dissociated hippocampal neurons in culture. This was blocked by pretreatment with the NMDAR antagonist, APV. This HGF effect was dependent upon elevations of intracellular Ca^{2+} and accompanying increases in autophosphorylation of CaMKII. Thus, Ca^{2+} -dependent processing underlies HGF's ability to increase dendritic arborization, and are consistent with the findings that increased hippocampal neuronal intracellular Ca^{2+} with Nle¹-AngIV treatment and facilitated hippocampal dendritic arborization.

8.14 Nle¹-AngIV, as well as shorter fragments, augmented hippocampal synaptic connectivity via the formation of new synapses. Functionality of these synapses was inferred from

analogue-induced spinogenesis, the colocalization of synaptic markers in newly-formed dendritic spines, coupled with enhanced mEPSCs.

8.15 Dihexa induces spinogenesis/synaptogenesis at pM concentrations, is orally-active, and slowly cleared from the blood (plasma stability half-life = 335.5min (5.6h)).

8.16 Dihexa crosses the BBB in sufficient quantity to facilitate spatial memory, as tested with the Scopolamine-induced amnesic rat model. 1.25-2mg/kg oral Dihexa blocked the impairing effects of ICV-administered Scopolamine. Dihexa-treated animals displayed an acquisition pattern of progressive improvement over 8d of training. In contrast, those receiving ICV Scopolamine without Dihexa performed poorly and persisted in requiring extended durations to locate the hidden pedestal. Dihexa treatment completely prevented Scopolamine-induced impairment, with Dihexa + Scopolamine-treated animals displaying equivalent learning curves to the group treated with neither. Next, probe trials were conducted on day 9 to evaluate the strength and persistence of the learned task. This involved removing the pedestal and requiring each animal to search during the entire trial duration of 2min. The group treated with 2mg/kg Dihexa followed by ICV Scopolamine spent significantly more time in the target quadrant than those treated only with Scopolamine, who had only chance-level target quadrant presence.

8.17 In a unilateral 6-OHDA-induced SN lesioned rat model of PD, rats were treated with 0.5mg/kg IP Dihexa on alternate days beginning 15d following surgery, continuing through day 48. Members of this group indicated significant reductions in walking stride length, particularly on the contralateral side and a 42% reduction in forepaw rope hang time. Following 3d of Dihexa treatment, rope hang times began improving and were soon equivalent with sham-operated animals and were significantly better than those of members with lesions but not Dihexa. Members of this latter group continued to show impaired rope hang times through day 48. Histological analysis revealed that animals treated with Dihexa had partial restoration of TH staining in the SN at 1mo following lesioning, and complete recovery by 3mo. It is hypothesized that Dihexa enhances dendritic sprouting, synaptogenesis, and neurogenesis. It is suspected that Dihexa acts to protect neurons from general growth factor deprivation and oxidative stress.

9.1 HGF mimics can act as mimetics (agonists) or antagonists. Mimetics act: to enhance cognitive function; as general neuroprotective/neuroregenerative agents; to facilitate wound repair; to improve insulin sensitivity and glucose transport; and to decrease tissue or organ fibrosis in order to prevent or reverse the symptoms of dementia, to protect from or reverse neurodegenerative disease, to facilitate repair of traumatic injury to the nervous system, to augment tissue and organ vascularization, to improve impaired wound healing, and to decrease or reverse fibrotic changes in organs like heart, lung, kidney, and liver. Antagonists act, for example, as anti-angiogenic and anti-cancer agents; to treat various malignancies and diseases like macular degeneration and diabetic retinopathy, which are associated with hypervascularization.

11.1 Dihexa improves cognition and increases synapses in rat models of cognitive dysfunction, but no studies have established its long-term safety, including potential tumorigenic and cancer metastatic effects.

11.2 In rat models of cognitive dysfunction Dihexa treatment improves cognition and increases spines/synapses. Dihexa did not improve cognitive functions in rats with normal cognition. There are no human studies to date. No studies have examined Dihexa for age-related diseases. Theoretically, Dihexa,, via HGF and c-Met activation could promote tumorigenesis and cancer

progression. No studies in animals or humans have examined long-term safety of Dihexa. Dihexa has a long half-life and there are theoretical concerns of c-Met activation leading to tumorigenesis and cancer progression.

11.3 In rats, Dihexa has been tested orally at up to 2mg/kg, ICV up to 1nM, IV up to 10mg/kg, and IP up to 20mg/kg.

11.4 Dihexa has a half-life of 12d following IV administration and 8.8d following IP administration in rats. Undetermined in humans.

11.5 Dihexa binds with high affinity to HGF and promotes c-Met signaling and HGF-dependent cellular activity. Aside from its essential role in embryonic development, activation of c-Met later in life can stimulate neurogenesis and protect against tissue insults in many types of cells, including neurons. These actions have been linked to improved cognitive functions and increased dendritic arborization, spinogenesis, and synaptogenesis. Of the AngIV analogues, Dihexa has been noted to be potentially the most promising for future testing in humans given the hydrophobicity, metabolic stability, BBB penetrance, and clinical data showing cognitive benefits.

11.6 Dihexa was originally developed by Joseph Harding, PhD, and his laboratory at Washington State University (WSU). M3 Biotechnology, a WSU spin-off company, was formed to further develop Dihexa and related compounds. Further development led to NDX-1017 (now called Fosgonimeton). M3 Biotechnology was renamed Athira Pharma in 2019 and NDX-1017 was renamed to ATH-1017.

11.7 In rat models of cognitive dysfunction, Dihexa treatment improves cognition and increases spines/synapses. Dihexa did not improve cognitive functions in rats with normal cognition.

11.8 In a systematic review of preclinical studies assessing cognitive effects of Renin-Angiotensin system peptides, 8/9 studies found AngIV and its analogues were effective in improving performance on spatial working memory and passive avoidance tasks. Dihexa has been studied in a few studies in rodents and cell culture. Based on a study in rats using [³H]Dihexa and [³H]Inulin infusions, Dihexa was found to penetrate the BBB and accumulate in the brain.

11.9 In rat hippocampal culture, Dihexa administration for 5d increased the number of spines by nearly 3x (41 spines/50um of dendrite with Dihexa, vs 15 spines with vehicle). An acute 30min application of Dihexa (1pM) also increased spines on neurons as well as the spine-head width, indicative of larger synapses implicated in memory functions. The mean spine-head width was 0.8um with Dihexa treatment and 0.67um for vehicle control. These newly-formed spines/synapses contained the same synaptic machinery as already-existing spines, as indicated by the presence of VGLUT1, Synapsin, and PSD-95. The synapses that were newly formed were also confirmed to be functional (based on a corresponding increase in AMPAR-mediated mEPSCs).

11.10 A follow-up study to the above-revealed that the cognitive and spine/synaptic effects of Dihexa were mediated by HGF/c-Met. Dihexa bound to HGF with high affinity, inhibited HGF dimerization, and acted synergistically with HGF to promote c-Met signaling, including cellular scattering, decreased cell adhesion, and increased cell motility and proliferation. Dihexa and HGF also acted synergistically to increase hippocampal neuronal spinogenesis.

11.11 The HGF antagonist (Hinge) delivered ICV blocked the pro-cognitive effects of orally-delivered Dihexa, as measured by the MWM test. It is worth highlighting that the HGF

antagonist alone, or Dihexa in the absence of Scopolamine, had no effect on cognitive performance, suggesting that the HGF/c-Met system is not engaged during normal learning under healthy conditions. The HGF/c-Met signaling has been shown to be upregulated in neurodegenerative diseases, injury, and stroke. It is thought that the HGF/c-Met system is designed to respond to injury and support synaptic plasticity and regeneration.

11.12 In a press release, the authors of the 2013 publication noted that Dihexa was 7 orders of magnitude more powerful than BDNF, based on cell culture assays of new neuronal connections. However, the effects of BDNF and Dihexa were not directly compared in the paper.

11.13 There have not been any studies that have tested Dihexa for age-related diseases. There is a theoretical concern for Dihexa's mechanism of action. Activation of HGF and c-Met is a key signaling pathway for many cancers. In cancer cells, HGF and c-Met are often overexpressed, and this overexpression correlates with tumorigenesis, metastasis, and poorer prognosis. No studies have tested the long-term safety of Dihexa treatment including its potential effects on tumorigenesis and cancer progression.

11.14 In preclinical studies, Dihexa has been used as one of the growth factor-free small molecules to differentiate hPSCs to hepatocyte-like cells. Dihexa is used during the hepatocyte maturation phase where hepatoblasts are differentiated to hepatocyte-like cells. In this phase, 100nM Dihexa, and 100nM Dexamethasone are administered for 10d.

11.15 In preclinical studies, Dihexa solutions were prepared by suspending Dihexa in DMSO at 1mg/mL, and subsequent serial dilutions in 50% DMSO or HPLC-grade H₂O. The rats received Dihexa dissolved in 75% DMSO IV (via a jugular vein or catheter), or IP. The typical injection volume was 200uL, yielding an estimated DMSO blood concentration of 0.46%. While DMSO is FDA-approved to treat certain conditions (e.g. painful bladder syndrome), DMSO can cause serious side effects when used in high concentrations.

11.16 In rats, Dihexa exhibited a long half-life of 12.68d (12d 16h 19min) following IV administration and 8.83d (8d 19h 55min) following IP administration. Dihexa had a serum half-life of 335.5min (5h 36min) in a study on rat serum.

11.17 There are several clinical trials testing Fosgonimeton, a phosphorylated form of Dihexa, testing its effects in mild-to-moderate AD and PD. Exclusion criteria for these trials include people with a history of or newly-diagnosed malignant tumor, epilepsy, myocardial infarction (MI) or unstable angina, clinically-significant cardiac issues (e.g. arrhythmia, cardiomyopathy), hypertension, hepatic impairment, and others.

12.1 In primary rat cortical neurons challenged with Abeta₁₋₄₂, Dihexa treatment significantly improved neuronal survival, protected neurite networks, and reduced Tau hyperphosphorylation. Cortical neurons treated with Dihexa exhibited a significant decrease in mitochondrial oxidative stress and cytochrome c release. Following Abeta injury, Dihexa significantly enhanced activation of pro-survival effectors ERK and Akt, and reduced activity of GSK-3beta, one of the main enzymes involved in Tau hyperphosphorylation. Dihexa also mitigated Abeta-induced deficits in Unc-5-like kinase 1 (ULK1) and Beclin-1, suggesting a potential effect on autophagy. Treatment with Dihexa protected cortical neurons from Glutamate excitotoxicity, and such effects were abolished in the presence of an Akt or MEK/ERK inhibitor. In vivo, Fosgonimeton administration led to functional improvement in an ICV Abeta₂₅₋₃₅ rat model of AD, as it significantly rescued cognitive function in the passive avoidance test. Thus, Fosgonimeton counteracts mechanisms of Abeta-induced toxicity.

12.2 Fosgonimeton is currently in clinical trials for mild-to-moderate AD.

12.3 Upon binding of HGF, the c-Met receptor undergoes dimerization and autophosphorylation, resulting in the activation of neurotrophic and pro-survival pathways including PI3K/Akt, ERK, and PKC pathways. Activation of c-Met by HGF plays a significant role in promoting development, maintenance, and repair of the nervous system. HGF signaling activates pro-survival and anti-apoptotic mechanisms, as well as modulation of neurotransmission. With regard to AD pathology, clinical data have demonstrated a decrease in the expression of c-Met in the brains of people with AD. Reduced HGF/c-Met signaling has been implicated in synaptic pathology observed in an AD mouse model. Correction of this deficit and enhancement of HGF/c-Met-induced neurotrophic and neuroprotective effects may present a novel therapeutic avenue for AD.

12.4 Fosgonimeton and Dihexa have demonstrated neurotrophic, neuroprotective, and anti-inflammatory effects in vitro and in vivo.

12.5 Dihexa promotes neuronal survival, preserves neurite networks, and reduces Tau hyperphosphorylation following Abeta₁₋₄₂ injury .

13.1 In one clinical trial, Fosgonimeton is being tested as an SC injection of 40mg or 70mg daily for mild-to-moderate AD.

14.1 There is an ongoing study of the long-term safety of daily SC injections of 40mg Fosgonimeton, with a planned time-frame of 48-173w.

15.1 An expression of concern has been published for the following article, so take it with a grain of salt. The expression of concern is not publicly available, so I cannot access it.

15.2 Dihexa exhibits excellent antidementia activity in the Scopolamine and aged rat models and marked synaptogenic activity.

15.3 Adult male Sprague-Dawley rats were administered 10mg/kg IV Dihexa and in vivo pharmacokinetics were determined. Dihexa exhibited rapidly decreased plasma levels from 0-4h, suggesting that both distribution and elimination occurred during this period. After 4h, the rate of clearance declined and plasma levels became more stable, exhibiting a relatively linear rate of decline, suggesting a role of pure elimination from 4-120h. The exception from the linearly-declining pattern of plasma levels was between 8.5-13h, when plasma levels were actually lower than at 24h. This suggests that a small fraction of Dihexa undergoes enterohepatic recirculation, which could cause an increase in plasma levels. Elevations in plasma concentrations due to enterohepatic recirculation are usually observed after a meal when bile containing Dihexa is released into the duodenum and is reabsorbed from the intestine. Since rats were allowed food 12h after the onset of the study, any enterohepatic recirculation would be expected to occur subsequent to 12h, the time period that corresponded with rising plasma levels of Dihexa.

15.4 Dihexa exhibited a long half-life of 12.68d (12d 16h 19min) following IV administration of 10mg/kg and similarly when delivered IP of 8.83d (8d 19h 55min). Dihexa appeared to be extensively distributed outside the central blood compartment and/or bound within the tissues as evidenced by its large volume of distribution (V_d). These results, which suggest Dihexa is very hydrophobic (logP), are in agreement with the outcome of QSAR modeling estimates generated by ADMET Predictor that calculated an Octanol/Water partition coefficient of 177.8 for Dihexa.

15.5 Not surprisingly because of its stability, hydrophobic character, and small size, Dihexa was predicted to be orally bioavailable. The predicted effective human jejunal permeability (P_{eff})

value for Dihexa (1.78) is intermediate between the predicted P_{eff} values for Enalapril (1.25) and Piroxicam (2.14), 2 orally-bioavailable drugs.

15.6 Dihexa was predicted to be 22.59% unbound to plasma proteins in circulation, thus making it available for distribution into the tissues.

15.7 In addition to its slow removal from the blood was a lack of Phase I metabolism. Phase I metabolism of Dihexa, which was determined using pooled male rat liver microsomes, was found to be very low, with an average intrinsic clearance (Cl_{int}) of 2.72uL/min/mg and an average half-life of 509.4min (8h 29min). To provide context for Dihexa's clearance rate, the stability of Piroxicam (60.2uL/min/mg), Verapamil (112.5uL/min/mg), and 7-Ethoxycoumarin (136.7uL/min/mg) was also monitored as high-, moderate-, and low-metabolized standards, respectively.

15.8 Dihexa exhibits pro-cognitive activity.

15.9 Dihexa induces spinogenesis in cultured hippocampal neurons. Actin-enriched spines increased in number in response to both Dihexa and Nle¹-AngIV following 5d of treatment at 1pM concentration that started on the 7th day in vitro (DIV7). The results revealed a near 3x increase in the number of spines stimulated by Dihexa and a >2x increase for Nle¹-AngIV. Both treatment groups differed significantly from the vehicle group for which the average number of spines/50um dendrite length was 15. The average number of spines for the Dihexa- and Nle¹-AngIV-treated groups was 41 and 32 spines/50um dendrite length, respectively.

15.10 The ICV MWM data with Dihexa indicate a modest but significant improvement in spatial learning performance even on the 1st day of testing, thus suggesting that the underlying mechanism responsible for the behavior must be rapidly engaged. Therefore, the ability of both Dihexa and Nle¹-AngIV to promote spinogenesis was assessed following an acute 30min application on the final day of culturing. The acute 30min application of Dihexa and Nle¹-AngIV, on DIV12, reveals a significant increase in spines compared with 30min vehicle-treated neurons (Dihexa mean spine numbers/50um dendrite length = 23.9; Nle¹-AngIV mean spine number/50um dendrite length = 22.6; vehicle control-treated neurons mean spine numbers/50um dendrite length = 17.4).

15.11 Strong correlations exist between spine size, persistence of spines, number of AMPARs, and synaptic efficacy. A correlation between the existence of long-term memories to spine-head volume has been suggested. Dihexa and Nle¹-AngIV both increased spine-head width. The mean spine-head width for Nle¹-AngIV was 0.87um, 0.8um for Dihexa, and 0.67um for vehicle controls.

15.12 Hippocampal neurons were treated with 1pM Dihexa or Nle¹-AngIV for 5d. Dihexa and Nle¹-AngIV significantly augmented dendritic spinogenesis in each of the 3 studies, with mean spine numbers for the combined studies for Nle¹-AngIV = 39.4, for Dihexa = 44.2, and for vehicle-treated neurons = 23.1. There was a correlation between newly-formed spines and synaptic markers VGLUT1, Synapsin, and PSD-95, indicating that the newly-formed spines contained the same synaptic machinery as already-present spines.

15.13 The mean frequency of AMPA-mediated mEPSCs recorded from vehicle-treated neurons was 3.06Hz from 33 cells, while Nle¹-AngIV induced a 1.7x increase (5.27Hz from 25 cells), and Dihexa produced a 1.6x increase (4.82Hz from 29 cells), confirming the expansion of functional synapses.

16.1 In hippocampal CA1 organotypic cultures, a better model than dissociated neurons, Dihexa and Nle¹-AngIV significantly augmented spinogenesis compared to vehicle. Spine numbers measured for control slices were 7 spines/50um dendrite length vs 11 spines/50um dendrite length for both Nle¹-AngIV and Dihexa-treated neurons.

17.1 This article has been retracted.

18.1 This is the notice of retraction for study 17.

18.2 Following an investigation by WSU, Fig 1B, 2A/C, and data in the subsequent erratum submission for the article have been found to contain falsified and/or fabricated data, and Leen H. Kawas and Joseph W. Harding were found to be solely responsible. Thus, I have chosen not to include data from study 17.

19.1 Dihexa increased the amount of AngIV present in mouse tissue in APP/PS1 mice relative to the WT group. Dihexa restored spatial learning and cognitive functions in the MWM test. Dihexa increased neuronal cells and the expression of Synaptophysin protein in APP/PS1 mice in Nissl staining. Dihexa decreased the activation of astrocytes and microglia, markedly reduced levels of the pro-inflammatory cytokines IL-1beta and TNFalpha and increased the levels of the anti-inflammatory cytokine IL-10. Dihexa activated the PI3K/Akt signaling pathway, while PI3K inhibitor Wortmannin significantly reversed the anti-inflammatory and anti-apoptotic effects of Dihexa on APP/PS1 mice.

19.2 AngIV has a cardioprotective effect against ischemia-reperfusion (R/I) injury by inhibiting apoptosis via the PI3K/Akt/mTOR pathway.

19.3 Dihexa at 1.44 and 2.88mg/kg decreased the impaired escape latency in the MWM of APP/PS1 mice. Dihexa decreased escape latencies to differing degrees, especially on the 4th and 5th days.

19.4 Dihexa downregulated GFAP and Iba-1.

19.5 Dihexa increased the number of neurons in the brains of APP/PS1 mice.

20.1 Dihexa is a promising candidate for further development as an adjunct therapy to promote limb functional recovery after surgical nerve repair, and has implications in peripheral nerve injury and limb transplantation.

21.1 7/11 studies of normal animals found AngIV had beneficial effects on tests of passive or conditioned avoidance and object recognition. In models of cognitive deficit, 8/9 studies found AngIV and its analogues (Nle¹-AngIV, Dihexa, LVV-Hemorphin-7 (LVV-H7)) improved performance on spatial working memory and passive avoidance tasks. 2/3 studies on Ang₁₋₇ found it benefited memory.

21.2 AngIV induces LTP via: VGCC-dependent Ca²⁺ neuronal influx, the activation of PKMzeta, a fragment of PKCzeta, which is linked to the maintenance of hippocampal LTP and various types of memory, and mediation by the DAergic system, given that doses of AngIV that improve learning in rats have also been found to increase stereotypic behaviors induced by DAergic agonists, effects which are reliant on DA receptor D₁ and D₂. Increases in DAergic neurotransmission following ICV Ang₁₋₇, which may be mediated by the action of AngIV or its derivative des-Phe⁶-AngIV, have also been reported.

21.3 ICV administration of either AngIV or LVV-H7 improved performance on a plus maze task relative to vehicle controls, with peptide-treated animals showing increased alternation scores. Similarly, in another study, Nle¹-AngIV or LVV-H7 was associated with significantly fewer errors, shorter distance traveled, and reduced latency in finding the escape tunnel, compared to the

controls, with differences observed in the first two measures as early as the 1st day of testing. The beneficial effects of Nle¹-AngIV were attenuated by coadministration of the AT₄ antagonist, Divalinal-AngIV.

21.4 Rats treated with Divalinal-AngIV delivered directly to the nucleus basalis magnocellularis (NBM) took significantly longer to find the hidden platform compared to vehicle controls, and performed at similar levels to rats treated with Scopolamine or Mecamylamine alone. Rats pretreated with Nicotine did not show these impairments. Carbachol was ineffective at alleviating the Divalinal-AngIV-induced impairment.

21.5 Nle¹-AngIV improved the rate of acquisition of the MWM, reduced the latency to find the pedestal and decreased swim distance.

21.6 ICV treatment with AngIV was associated with better object recognition memory, whether given 5, 10, or 15min prior to testing.

21.7 After SC injection of AngIV, object recognition was improved. Different strains of mice responded differently to increasing AngIV doses. AngIV enhanced recall at 4.7 and 47ug/kg in DBA2 mice, while C57 strain mice only showed improvement at the lower dose of 4.7ug/kg, and CD mice improved with neither dose.

21.8 AngIV at doses of 100pM, 1nM, and 10nM, but not 10pM, increased the percentage of time spent exploring the novel object in a truncated version of the NOR task.

22.1 HGF-induced c-Met activation exerts beneficial neuroprotective effects in adulthood, specifically in neurodegenerative disorders, and in preclinical models of cerebral ischemia, spinal cord injuries, and neurological pathologies, such as AD, ALS, and multiple sclerosis (MS). HGF is a key factor in preventing neuronal death and promoting survival through pro-angiogenic, anti-inflammatory, and immunomodulatory mechanisms. HGF acts on neural stem cells (NSCs) to enhance neuroregeneration.

22.2 HGF is a pleiotropic cytokine isolated as a potent cell motility factor for epithelial cells and as a strong mitogen for hepatocytes in primary cultures.

22.3 The expression and functions of HGF/c-Met are fundamental during embryogenesis. In HGF- or c-Met-null mice, the development of the liver is severely impaired and the placental labyrinth is hypomorphic, leading to in utero death. Moreover, ablation of these genes leads to the complete absence of hypaxial muscles (diaphragm, limbs, tongue), reflecting that HGF/c-Met have a pivotal role in proliferation and motility of these muscle progenitors. HGF or c-Met ablation affects the wiring of the nervous system, leads to a decreased survival of sensory and sympathetic neurons, and reduces axon bundling of certain motor nerves.

22.4 c-Met agonism is involved in angiogenesis, wound healing, cell scattering, proliferation, and cancer invasion.

22.5 HGF has pro-angiogenic, anti-fibrotic, anti-inflammatory, immunomodulatory, and neuroregenerative properties.

22.6 Upon HGF binding to c-Met, the receptor is activated by dimerization. The activation occurs via trans-phosphorylation of 2 catalytic tyrosines (Y1234 and Y1235) located in the kinase domain, followed by the subsequent autophosphorylation of the 2 docking sites (Y1349 and Y1356) in the C-terminal tail. The multifunctional docking site of c-Met recruits to the plasma membrane stereotypical adaptor/signaling molecules such as GAB1, GRB2, SPH2, PI3K, Src, PLCgamma, SHC, and STAT3. These trigger downstream signaling pathways, including Akt, Ras, and STAT3 pathways, which are required for c-Met biological activity. Akt activates mTOR,

promoting protein synthesis and cell growth. Ras leads to the activation of MAPK cascades, including ERK, p38 MAPK, and JNK. Moreover, Ras induces Rac1, which in turn leads to further stimulation of p38 MAPK and JNK. ERK is mainly implicated in cell proliferation and Rac1 in cell migration, while p38 MAPK and JNK are involved in differentiation and apoptosis. STAT3 activation is essential for tubulogenesis.

22.7 HGF is a neurotrophic factor and an axonal chemoattractant in embryonic development.

22.8 Mouse embryos possessing non-functional c-Met result in short, less branched spinal sensory nerves and apoptotic sympathetic neuroblasts in vivo.

22.9 HGF cooperates with NGF in vitro to enhance the axonal growth of dorsal root ganglion (DRG) sensory neurons and sympathetic neurons.

22.10 HGF is implicated in guiding spinal motoneurons to their target muscles in the limbs.

22.11 HGF promotes survival and axon outgrowth of specific subpopulations of motoneurons and guides cranial and vagus motor axons to the pharyngeal arches.

22.12 Studies in mice show that HGF and c-Met expression is detectable in the brain during early embryonic development at embryonic day 12 and 13 (E12 and E13), respectively.

Expression of both HGF and c-Met is first found in the cortical ventricular zone and later in the cortical plate. HGF is also expressed in the proliferative zone of the ganglionic eminence. Mice with inactivation of the urokinase plasminogen activator receptor (uPAR) gene, a key component of HGF activation, show diminished levels of HGF in cortical interneurons beginning at E16.5, indirectly suggesting that HGF has motogenic effects also on cortical interneuron precursors. Olfactory interneuron precursors are chemoattracted by HGF from the subventricular zone (SVZ) of the lateral ventricles, through the rostral migratory system (RMS), to the olfactory bulb. HGF and c-Met are simultaneously expressed during cerebellum development, which occurs predominantly postnatally. HGF increases the proliferation of cerebellum granule cell precursors and this effect is reduced in a hypomorphic c-Met mutant (Grb2-binding incompetent) that leads to a smaller size of the cerebellum.

22.13 Specific neurotrophic effects of HGF were reported on Calbindin-D-containing hippocampal neurons and DAergic mesencephalic neurons.

22.14 Sites of HGF and c-Met expression in the postnatal brain include the CA1 area of the hippocampus and limited regions of the cortex. c-Met transcript levels in the cerebral cortex are quite low at prenatal stage and increase significantly starting from postnatal day P0 to peak at P17. Since then, c-Met levels remain elevated in the neocortex, hippocampus, and subcortical limbic regions of mice forebrains during the 2nd postnatal week, decreasing from P21 to adult age. Interestingly, there is a coincidence between c-Met peak expression and the early and late stages of neurodevelopmental processes such as dendritic morphology, neurite outgrowth, and synaptogenesis. Exogenous HGF enhances cortical pyramidal dendrite arborization in organotypic slice cultures of mouse somatosensory cortex at P6 and P7. c-Met expression can be detected at a subcellular level in the growth-cone of P0 mice developing cortical tissue as well as in cultured hippocampal neurons, suggesting a functional role mediated by c-Met in neuronal growth. c-Met is enriched, during forebrain formation, in developing axon tracts and in the neuropil, mostly in pre- and postsynaptic compartments, mirroring an active transport to the synapse. The c-Met transcript is also detected in the neuropil. Despite that, phospho-c-Met (p-c-Met) is not detected in axon tracts, while it is enriched in the neuropil. c-Met expression levels in mice brain preparations of the striatum radiatum layer of hippocampal CA1 region are

comparable in pre- and postsynaptic terminals at P7 peak in mice. At a later stage, c-Met is predominantly presynaptic, reflecting a dynamic regulation of c-Met expression depending on the postnatal age of the examined mice. The p-c-Met protein levels peak at P7-P14 in the neocortex, a period corresponding to synaptogenesis and dendritic outgrowth, and are downregulated during the 3rd postnatal week, between P14 and P17, when pruning and functional synapse maturation occur.

22.15 c-Met forms clusters at excitatory synapses in relatively mature cultured hippocampal neurons and colocalizes with the postsynaptic protein PSD-95 and with the glutamatergic NMDAR subunit GluN2B.

22.16 c-Met is enriched in the postsynaptic density fraction, and HGF administration induces c-Met phosphorylation and clustering of synaptic proteins such as GluN2B, CaMKII, and the AMPAR GluA1, suggesting a functional connection between glutamatergic synapses and c-Met signaling. c-Met activation is thus involved in neuroplasticity, as suggested by the fact that HGF administration to hippocampal slices enhances the phosphorylation of the NMDAR subunit GluN1, augments the NMDAR-mediated currents, and increases the amplitude of LTP induced by elevated neuronal activity. The HGF/c-Met system thus contributes to the activity-dependent regulation of physiological learning and memory performance in the adult brain.

22.17 c-Met signaling enhancement in primary hippocampal neurons during the 2nd culture week leads to dendritic growth, while at a later developmental stage, c-Met signaling leads to dendritic spine development. These observations, combined with the fact that spine head size and geometry are associated with AMPAR content and maturation status, suggest that c-Met signaling may control many aspects of glutamatergic synapse development, including the timing of excitatory synapse maturation. Conditional knockout (cKO) of c-Met in forebrain excitatory neurons reduces dendritic complexity in mice and correlates with an earlier functional maturation of hippocampal CA3 > CA1 synapse and a disrupted intracortical connectivity phenotype in frontal cortex. At a molecular level, c-Met cKO mice showed altered glutamate receptor expression profiles in the hippocampus at P14, including increased GluA1, GluN2A, and decreased GluN2B at the synaptic sites.

22.18 HGF-induced synapse formation is mediated by PI3K and not MAPK activation, as inhibition of the PI3K/Akt pathway affects synaptogenesis, while blocking MAPK/ERK does not decrease synapse density and only affects dendritic growth.

22.19 c-Met activation leads to stimulation of Rho family small GTPase cdc42 in a PI3K-dependent manner. Rho small GTPase family (Rac1, cdc42, RhoA) has an established role in mediating neurite growth and spine morphogenesis. c-Met-mediated cdc42 activation is associated with enhanced dendritic growth and spine morphogenesis.

22.20 c-Met has an important role in neuron architecture construction and synapse maturation. The reduced expression or deletion of c-Met in vivo in mice results in altered synapse maturation.

22.21 c-Met is expressed in brain areas involved in higher levels of cognition, executive functions, and language skills.

22.22 The c-Met "C" allele is enriched in the autism spectrum disorder (ASD) subpopulation with impaired social interactions. The C allele reduced expression of c-Met. The C allele is also enriched in a subpopulation of ASD with GI dysfunction. Structural and functional connectivity

are reduced by the C/C genotype. Thus, the C allele of c-Met is considered a risk factor for ASD.

22.23 c-Met levels in postmortem temporal lobe samples of ASD or healthy individuals showed a reduction in c-Met transcripts and proteins in ASD compared to controls.

22.24 Low levels of HGF are found in the plasma of ASD patients.

22.25 The rs1858830 c-Met promoter variant is predictive of atypical fMRI activation and deactivation pattern of the human brain to social stimuli, and it impacts functional and structural connectivity in the temporo-parietal lobes, which selectively express c-Met.

22.26 Forebrain-specific c-Met cKO mice show altered circuit formation and function including synaptic plasticity.

22.27 The upregulation of the HGF gene is implicated in schizophrenia. HGF was found to be upregulated in fibroblasts collected from a unique population of schizophrenia patients in northern Sweden.

22.28 The expression of both HGF and c-Met receptor mRNA is strongly increased in response to cerebral ischemic injury. HGF improves the neurological consequences of stroke by reducing infarct volume of the ischemic brain and by preventing neuronal death via the reduction of apoptosis.

22.29 HGF's anti-apoptotic effects are mediated by the PI3K/Akt pathway in rat cerebellar granule cells in vitro in an ERK-independent manner.

22.30 In cortical neurons subjected to hypoxia/reoxygenation, HGF protects from apoptosis via the ERK pathway and, to a lesser extent, the PI3K/Akt pathway.

22.31 In ischemic brain injury in vivo, HGF inhibits apoptosis via transient activation of ERK. The induction of neuroprotective genes, such as Bcl-2 and Bcl-xl, also contributed to the inhibition of apoptosis by HGF, and is promoted by STAT3 activation. The implication of the c-Met/STAT3/Bcl-2 pathway in HGF protection from apoptotic neuronal death in stroke has recently been suggested in a rat model of transient middle cerebral artery occlusion (tMCAO).

22.32 HGF reduces a caspase-independent ischemic neuronal death by preventing apoptosis-inducing factor (AIF) translocation downstream of poly(ADP-ribose)polymerase-1 (PARP1) and p53.

22.33 Oxidative DNA stress occurring after cerebral ischemia is associated with a decrease in apurinic/apyrimidinic endonuclease/redox factor-1 (APE/Ref-1) expression. APE/Ref-1 is a multifunctional protein in the DNA base repair pathway that is responsible for repairing apurinic/apyrimidinic sites in DNA after oxidation. HGF counteracts the oxidative DNA damage after cerebral ischemia by increasing the level of APE/Ref-1+ cells in the hippocampal CA1 region. Also in the CA1 region, after cerebral ischemia, HGF decreases the activity of NADPH oxidase and consequent ROS generation in glia-like cells. In line with this, HGF decreases oxidative stress in an AD Tg mouse model. p-CREB, mediated by PI3K and MAPK signaling pathways, that are known to decrease neurotoxicity and oxidative stress-induced neuronal death, may be implicated.

22.34 HGF protects neurons in tMCAO rats via inhibition of apoptosis and autophagy. Mechanistically, neuronal death protection from autophagy seems to involve the c-Met downstream mTOR signaling, whose activation (phosphorylation) is enhanced after tMCAO.

22.35 HGF reduces NMDA-mediated neuronal death in cerebellar granule neurons in vitro. HGF prevents NMDA-induced neuronal death in hippocampal neuron cultures in vitro by blocking pro-apoptotic caspase-3 activation and translocation of AIF into the nucleus.

22.36 In ALS, HGF treatment prevents excitotoxic insult by enhancing the expression of glutamate transporter-1 (GLT-1) in astrocytes and subsequent clearance of excessive glutamate.

22.37 Exogenous HGF treatment shows anti-fibrotic activity in animal models, such as the tMCAO model, exerted through the decrease of glial scar formation and scar thickness of the brain pia matter. HGF mostly exerts an anti-fibrotic effect on glial scar reduction by inhibiting the synthesis of the proteoglycans produced by the activated astrocytes through the complete block of TGF-beta1 and TGF-beta2 action. The astrocyte activation and the glial scar formation were inhibited in an SCI model transplanted with mesenchymal stem cells (MSCs) overexpressing HGF. The Tg overexpression of HGF in the nervous system attenuates spinal motoneuronal death and axonal degeneration, delays onset of the disease, and prolongs the lifespan of SOD1-G93A mice. The enhancement of c-Met levels in neurons selectively delays the disease onset in ALS mouse models, prolonging lifespan, slowing motor neuron loss, and ameliorating motor performance. HGF also reduces the number of astrocytes by retaining the levels of the glial-specific glutamate transporter-1, thus favoring a reduction in excitotoxicity. HGF overexpression attenuates monocyte chemoattractant protein-1 (MCP-1) induction, predominantly in astrocytes, thus reducing accumulation of activated microglia.

22.38 HGF induces angiogenesis in the nervous system.

22.39 HGF was originally identified as a potent mitogen for hepatocytes. It has also been recognized as a member of powerful pro-angiogenic growth factors.

22.40 HGF induces DNA synthesis and proliferation of vascular endothelial cells through the activation of ERK and STAT3. HGF protects endothelial cells from hypoxic death by the Akt pathway and p38 MAPK, as well as via the enhancement of Bcl-2 expression.

22.41 HGF is also expressed in neuro-microvascular endothelial cells, and an association between growth factors and increased angiogenesis in ischemic injuries is reported in the literature.

22.42 HGF amplifies angiogenesis after tMCAO in rats and reduces apoptosis in vitro.

22.43 HGF protects endothelial cells against apoptosis in cerebral ischemia animal models and prevents the associated learning and memory dysfunctions. HGF can also protect the integrity of the BBB. HGF protects BBB disruption after microsphere embolism (ME)-induced sustained cerebral ischemia by avoiding the inhibition of Bcl-2, thus decreasing apoptosis, and by reducing the expression of occludin and zonula occludens-1 (ZO-1), which are tight junction proteins, in cerebrovascular endothelial cells.

22.44 A pathway implicated in HGF's angiogenic roles includes a strong association with the E-twenty-six (Ets) pathway, which is itself implicated in the regulation of developmental and mitogenic signals. HGF can upregulate transcription factor Ets-1, thus enhancing the expression of VEGFs. HGF and VEGF play a synergistic role in promoting endothelial cell survival by enhancing the anti-apoptotic genes Bcl-2 and A1. HGF downregulates the expression of Thrombospondin 1, which negatively regulates angiogenesis.

22.45 HGF regulates the inflammatory and immune response in many pathological conditions by modulating cell migration, antigen presentation, T-cell effector function, and cytokine production, thus exerting a protective function.

22.46 HGF decreases NF-kappaB activation to downregulate adhesion molecules such as ICAM-1/E-selectin on the endothelial surface, thus inhibiting neutrophil infiltration and suppressing ischemia-related injury in various organs. HGF is able to directly target activated eosinophils through the downregulation of inflammatory mediators during allergic responses.

22.47 HGF ameliorates tissue damage and functional recovery after the acute phase in a mouse model of SCI by exerting anti-inflammatory effects through the diminution of LPS-activated macrophages and neutrophil infiltration. Mechanistically, HGF inhibits inflammatory M1-like macrophage production of pro-inflammatory cytokines and chemokines (such as IL-1beta, IL-6, TNF-alpha, CXCL1, CXCL10, and CCL2) by disrupting the NF-kappaB signaling through the enhancement of heme oxygenase-1 (HO-1) transcriptional pathway and the inactivation of the GSK-3beta activity.

22.48 HGF stimulates macrophage differentiation in dendritic immunosuppressive cells, thus facilitating the induction of T-lymphocyte regulators, associated with a pivotal role in anti-inflammation. HGF can mediate neuroinflammation in the context of neurodegenerative autoimmune diseases such as MS, leading to an indirect effect on the immune system. HGF exerted an anti-inflammatory effect through the generation of tolerogenic dendritic cells with the consequent suppression of auto-reactive T-helper 1 cells (Th1) and Th17 cells and leading to the reduction of CD4+ T-cell-mediated nervous system injury in the experimental autoimmune encephalitis (EAE) animal model of MS, in which HGF is overexpressed. The administration of HGF or adoptive transfer of HGF-treated dendritic cells in recipient EAE mice reduces disease progression and induces dendritic tolerance. There was an increase in the frequency of both PNS and CNS anti-inflammatory IL-10-secreting regulatory T-cells (Treg cells) and, in parallel, a decrease of pro-inflammatory Th1 and Th17 responses.

22.49 HGF stimulated the development of neurons and the proliferation and migration of myelinating oligodendrocytes (OLs), reduced pro-inflammatory cytokines, and enhanced anti-inflammatory cytokines.

22.50 HGF promoted re-myelination in vitro.

22.51 HGF contributes to the maintenance of self-renewal and proliferation of neural stem cells (NSCs) in the SVZ, which express c-Met and produce HGF. HGF promotes self-renewal and expansion of NSCs forming neurospheres. Mechanistically, the mitogenic effect promoted by HGF in SVZ cells is dependent on the activation of the ERK pathway.

22.52 In the context of SCI recovery, there is a positive effect of HGF in the promotion of grafted NSC differentiation and synapse formation between new neurons and the descending corticospinal fibers.

22.53 HGF inhibits the BMP/Smad pathway and promotes differentiation of NSC neurospheres into neurons and axonal outgrowth in vitro similarly to bone marrow stem cells (BMSCs) and in a dose-dependent fashion. HGF promotes differentiation of BMSCs into neuron-like cells in the context of SCI, in combination with GDNF.

22.54 hUC-MSCs infected with Adenovirus carrying the HGF gene promotes a better regeneration of DAergic neurons damaged by MPP⁺ (PD model) compared to hUC-MSCs alone.

26.1 Dihexa was not effective against dysfunction in a 3-NP model of HD.

28.1 Dihexa has been shown through immunocytochemistry to be the most potent positive modulator of HGF/c-Met among the screened candidates. Primary hippocampal neurons treated with Dihexa showed enhanced synaptogenesis and neurite outgrowth, supporting the neurotrophic effects of positive modulators of the HGF/c-Met system. Dihexa protects against neurotoxic insults in primary cortical neuron cultures. In vivo, treatment with Dihexa rescued cognitive deficits in the rat Scopolamine-induced amnesia model of AD.

28.2 Dihexa + HGF exhibited a significant increase in levels of p-ERK and p-Akt.

28.3 Dihexa, in combination with a sub-effective dose of HGF, resulted in significant cell scattering at doses of 10pM, 100pM, and 1nM, each compared to HGF alone, suggesting that Dihexa drives HGF/c-Met-dependent cellular behaviors.

28.4 Hippocampal neurons treated with 10nM Dihexa exhibited a significant increase in neurite length per cell compared with vehicle. Cotreatment with 1nM Dihexa and 5ng/mL HGF also significantly increased neurite length per cell, whereas HGF alone showed no significant differences compared with vehicle. There was a significant effect of Dihexa on the number of branch points per cell.

28.5 Treatment with 1nM Dihexa, and 1nM Dihexa alongside 5ng/mL HGF, both significantly enhanced the immunofluorescence intensity of Synaptobrevin-II compared with vehicle. Cotreatment with 1nM Dihexa and 5ng/mL HGF significantly increased the number of synapses compared with vehicle. Thus, Dihexa can induce potent neurotrophic effects to facilitate development of neuronal projections and formation of new synapses.

28.6 Activation of the HGF/c-Met system counteracts neuronal damage induced by glutamate toxicity, oxidative stress, and ischemic injury. HGF/c-Met signaling exerts antiapoptotic effects via activation of ERK and/or Akt and supports expression of neuroprotective genes Bcl-2 and Bcl-xl.

28.7 Cortical neurons treated with Dihexa exhibited significant improvement of cell viability against neuronal damage induced by MPP⁺, glutamate, LPS, and H₂O₂. Treatment with 1uM Dihexa rescued cell viability against neurological insults with the following percentages of recovery: 85% vs MPP⁺, 61% vs glutamate, 93% vs LPS, and 144% vs H₂O₂. Significant improvement in cell viability was observed at all tested concentrations (1nM-1uM) of Dihexa against all insults.

28.9 Dihexa rescues Scopolamine-induced learning deficits.

28.10 HGF signaling is known to promote NMDAR activation. Overexpression of HGF in the CNS of rodents improves performance in the MWM, likely mediated in part by NMDAR modulation.

28.11 SC delivery of 1mg/kg Dihexa led to reduced escape latency over time in the MWM of rodents impaired by Scopolamine, who otherwise did not improve their escape latency over training days. Dihexa significantly improved the frequency of entering the platform zone, percentage of time spent in the platform zone, and mean distance from the center of the platform compared with vehicle-treated Scopolamine animals in the probe trial conducted 24h after training completion. Animals receiving Dihexa+Scopolamine demonstrated successful retention of the MWM task. Thus, Dihexa delivered SC is neuroactive and significantly improves spatial learning deficits from a Scopolamine challenge.

28.12 Dihexa has poor stability when exposed to the digestive enzymes present in gastric fluid and intestinal fluids, with only 7.8% and 0.1% remaining after 1h of exposure to simulated

gastric fluid and intestinal fluid, respectively, and ultimately has poor oral bioavailability with high variability in exposure. Thus, Dihexa is unlikely to be viable as an oral drug. The maximum aqueous solubility of Dihexa is <50ug/mL, which limits the use of other clinically-viable routes of administration, including SC injection. In vivo activity studies using Dihexa required a pure DMSO vehicle to achieve reasonable dose volumes.

28.13 Modification of Dihexa to improve solubility led to the creation of Fosgonimeton, a phosphorylated prodrug to Dihexa, which is soluble at concentrations >70mg/mL in aqueous solutions. After SC delivery, Fosgonimeton was rapidly metabolized to Dihexa, with Dihexa appearing by the first tested time points: 5min in rats and 15min in mice. Delivery of Fosgonimeton results in higher systemic exposure and an almost 4x higher C_{max} in rats compared to the equivalent direct dose of Dihexa, with a Dihexa concentration of 1.198ug/mL after Fosgonimeton, after metabolism, and only 329ng/mL after direct delivery of Dihexa. CNS penetration and distribution of Fosgonimeton and Dihexa were measured after SC injection of Fosgonimeton. Dihexa is distributed to all regions with a T_{max} of 10min and an exposure range of 23.75-58.3ng/g of tissue. Fosgonimeton itself was undetectable in all brain regions, suggesting it did not penetrate the BBB per se. CNS delivery of Fosgonimeton is therefore dependent on the activity of Dihexa. Fosgonimeton physicochemical characteristics are substantially improved compared to Dihexa, resulting in superior PK properties.

28.14 After 12mg/kg SC injection of Fosgonimeton, Dihexa was found at the following concentrations in the following brain regions:

- Whole brain 31.9ng/g
- Striatum 24.1ng/g
- Olfactory bulb 43.7ng/g
- Hippocampus 49.8ng/g
- Cerebellum 23.8ng/g
- Cerebral cortex 36.5ng/g
- Brainstem 58.3ng/g

28.15 After 5mg/kg SC injection of Fosgonimeton, Dihexa was found in the whole brain at 7.9ng/g.

28.16 2w of SC injection of Fosgonimeton prevents LPS-induced cognitive deficits in vivo at all tested doses (0.25mg/kg, 0.5mg/kg, 1mg/kg, and 1.25mg/kg). Percentage of recovery was calculated at 61.76% for the 0.25mg/kg group, 79.4% for 0.5mg/kg, 64.7% for 1mg/kg, and 64.7% for 1.25mg/kg. Memantine showed a 76.4% recovery rate.

28.17 LPS-challenged THP-1 macrophages treated with 1uM Dihexa exhibited a significant decrease in levels of IL-1beta release, TNF-alpha release, and IL-6 release compared to vehicle.

28.18 Dihexa significantly enhanced activation of ERK and Akt pathways in the presence of a subthreshold concentration of HGF, implying that positive modulation of HGF/c-Met by Dihexa is sufficient to activate the entire downstream HGF/c-Met pathway.

28.19 MDCK cell scattering, known to be activated by c-Met, was enhanced by treatment with Dihexa, confirming that activation of c-Met and its downstream effectors can have effects on cellular behaviors.

28.20 Dihexa significantly enhanced synaptogenesis, synaptic vesicle clustering (a surrogate measure of synaptic strength), and induced neurotrophic effects similar to those of exogenous HGF treatment in AD-relevant tissues.

28.21 HGF treatment of neuronal cultures modulates NMDAR activity through activation of effector proteins, including PKC, which phosphorylates NR1 subunits of the NMDAR and modulates surface expression, synaptic localization, and activation of the NMDAR. Increased synaptic localization of the NMDAR and augmentation of transreceptor currents may bolster LTP and related memory formation.

28.22 Fosgonimeton, Dihexa, and other PAMs of the HGF/c-Met system engage the following mechanisms: stimulates neural stem cell (NSC) proliferation, differentiation, and neurogenesis; induces neurite outgrowth and synaptogenesis; supports healthy neuronal networks; promotes cell survival and resistance to damage; modulates neuroinflammation; and promotes clustering and activity of NMDARs at the synaptic cleft to enhance pro-cognitive NMDAR activity, including LTP, while preventing excitotoxicity.

29.1 The AngIV analogue Norleual [Nle-Tyr-Ile-Psi-(CH₂-NH₂)³⁻⁴-His-Pro-Phe] exhibits structural homology with the hinge (linker) region of HGF and is hypothesized to act as a hinge region mimic. Norleual competitively inhibited the binding of HGF to its receptor c-Met in mouse liver membranes, with an IC₅₀ of 3pM. Norleual was able to inhibit HGF-dependent signaling, proliferation, migration, and invasion in multiple cell types at pM concentrations. Ex vivo studies demonstrated that Norleual exhibited potent antiangiogenic activity, an attribute that would be predicted of an HGF/c-Met antagonist. Norleual suppressed pulmonary colonization by B16-F10 murine melanoma cells, which are characterized by an overactive HGF/c-Met system.

30.1 Retracted.

31.1 Divalinal-AngIV (Divalinal), an HGF/c-Met-antagonistic AngIV analogue, disrupted the acquisition of Methamphetamine-induced dependence in rats as measured by conditioned place preference (CPP). Acute and chronic ICV infusion of Divalinal prior to each daily SC injection of Methamphetamine prevented acquisition. Divalinal was unable to prevent Methamphetamine-induced reinstatement after prior acquisition followed by extinction trials.

32.1 Retracted.

35.1 An autism spectrum disorder (ASD)-associated polymorphism disrupts c-Met gene transcription, and there are reduced levels of c-Met protein expression in the mature temporal cortex of subjects with ASD.

35.2 Gene association studies identified 2 common risk alleles, one of which negatively regulates c-Met gene transcription, consistent with a 2x reduction in c-Met expression in postmortem temporal cortex of subjects with ASD. Overrepresented copy number variations (CNVs) that delete c-Met were reported in one study, suggesting multiple genetic mechanisms for disrupting c-Met in ASD. Signaling pathways that c-Met is known to interact with contain at least 8 members that also show association with ASD.

36.1 c-Met is a potent modulator of key social brain circuitry in children and adolescents with and without ASD. c-Met risk genotype predicted atypical fMRI activation and deactivation patterns to social stimuli (i.e. emotional faces), as well as reduced functional and structural connectivity in temporo-parietal regions known to have high c-Met expression, particularly within the default mode network (DMN). These effects were more pronounced in individuals with ASD.

36.2 The c-Met variant MET rs1858830 (the “C” allele) is associated with risk of autism. The C allele correlates with lower levels of c-Met transcript and protein expression. This common variant may increase risk but is not “disorder-causing”. The rs1858830 C allele moderates the severity of social symptoms in ASD, whereby individuals with ASD who carry the risk allele have more severe social and communication phenotypes than those who do not. Individuals can carry 0, 1, or 2 copies of the C allele. G is the other allele. The C allele is enriched in individuals with ASD, particularly multiplex families and in the most highly impaired individuals with ASD. C allele carriers (the “risk” group) displayed a pattern of hyperactivation and reduced deactivation in specific regions in which c-Met is expressed in primates and humans. The risk and non-risk (G allele carriers) both activated primary/secondary visual cortices, thalamus, and amygdala; however, the risk group activated amygdala and striatum more robustly than the non-risk group. The non-risk group displayed widespread deactivation (i.e. reduced activity while viewing faces vs fixation crosses). The deactivation was most prominently displayed in midline structures of the DMN including the PCC and perisylvian regions centered on the primary auditory cortex. In contrast, the intermediate risk group did not deactivate, but not to the same extent as the non-risk group, and the risk group appeared to show slight activation in these regions on average.

36.3 Neocortical pyramidal neurons in c-Met-cKO mice exhibited altered dendritic architecture and increased spine head volume, as well as a concomitant increase in local intralaminar excitatory drive onto corticostriatal neurons. This heightened local-circuit connectivity is highly relevant to ASD risk and the current hypothesis regarding increased local circuit connectivity and decreased long-range connectivity of brain networks in individuals with ASD.

36.4 The c-Met risk genotype significantly modulated functional connectivity, such that those in the highest risk group (CC) had reduced intrinsic connectivity between the PCC and mPFC as well as other nearby regions in the PCC compared to the non-risk group (GG).

36.5 The c-Met risk genotype predicted marked reductions in fractional anisotropy (FA) across a restricted number of major white matter tracts known to connect the very same regions implicated in the above functional connectivity analyses. Compared to non-risk allele homozygotes displayed lower FA in multiple major tracts in temporo-parieto-occipital regions that exhibit high c-Met expression developmentally (i.e. splenium of the corpus callosum, superior/inferior longitudinal fasciculus, and cingulum).

36.6 ASD patients with the CG or CC genotype had significantly higher social impairment.

36.7 There is a direct relationship between c-Met risk genotype and increased symptom severity within individuals with ASD. The c-Met risk genotype may contribute to both alterations in brain circuitry and disrupted social behavior.

36.8 The C allele of c-Met alters functional activity patterns to social stimuli, DMN functional connectivity, and white matter integrity.

36.9 The presence of a c-Met C allele was associated with much stronger disruptions in white matter integrity than having an ASD diagnosis.

37.1 Schizophrenic patients have 3.81x higher HGF levels than non-schizophrenic individuals.

37.2 HGF is a key factor in proper migration of interneurons from ganglionic eminence to cerebral cortex.

39.1 c-Met signaling regulates the growth and development of axons and may contribute to the wiring of cortical and limbic circuits in rodent forebrains.

39.2 In the macaque, robust c-Met expression was largely restricted to the PCC, inferior temporal, posterior parietal, and visual cortices, including face-processing regions.

40.1 Fosgonimeton was administered 1x/d via SC injection to 88 subjects. Single dose studies were performed on healthy young men, and multiple-ascending dose studies were performed in elderly subjects. There was a fixed-dose study performed on AD patients. The single doses used were 2, 6, 20, 40, 60, and 90mg. The multiple ascending doses used were 20, 40, 60, and 80mg, for 9d. The fixed-dose study involved administration of 40mg SC Fosgonimeton for 9d. There was no accumulation over 9d. The main effect of Fosgonimeton on qEEG was acute and sustained Gamma power induction. In AD subjects, there was a significant effect toward ERP P300 latency normalization compared to placebo.

40.2 After SC injection, Fosgonimeton was rapidly absorbed ($T_{max} \sim 15\text{min}$) and converted in plasma to Dihexa ($T_{max} \sim 30\text{min}$). Both Fosgonimeton and Dihexa were rapidly eliminated from plasma with a half-life of $\sim 20\text{min}$ and 1.5h, respectively. A terminal elimination phase for Dihexa with a half-life of 5h was occasionally observed at doses $\geq 40\text{mg}$ when plasma concentrations dropped to $\sim 1\%$ of the C_{max} .

40.3 After SC injection, Fosgonimeton showed linear pharmacokinetics over a dose range of 2-90mg.

Studies:

[US8598118B2 - Hepatocyte growth factor mimics as therapeutic agents - Google Patents](#) 1

[Facilitation of Hippocampal Synaptogenesis and Spatial Memory by C-Terminal Truncated Nle1-Angiotensin IV Analogs - PMC](#) 2

<https://www.alzforum.org/therapeutics/fosgonimeton> 3

[Hepatocyte growth factor mimetic protects lateral line hair cells from aminoglycoside exposure - PMC](#) 4

<https://sci-hub.red/https://pubmed.ncbi.nlm.nih.gov/25649658/> 5

https://journals.sagepub.com/doi/10.3233/JHD-231507?url_ver=Z39.88-2003&rfr_id=ori:rid:crosref.org&rfr_dat=cr_pub%20%200pubmed 6

[Efficiently generate functional hepatic cells from human pluripotent stem cells by complete small-molecule strategy - PMC](#) 7

<https://sci-hub.red/https://www.sciencedirect.com/science/article/abs/pii/S0301008214001245?via%3Dihub> 8

[US8598118B2 - Hepatocyte growth factor mimics as therapeutic agents - Google Patents](#) 9

[Small-Molecule-Driven Hepatocyte Differentiation of Human Pluripotent Stem Cells - PMC](#) 10

https://www.alzdiscovery.org/uploads/cognitive_vitality_media/Dihexa_1.pdf 11

[Fosgonimeton attenuates amyloid-beta toxicity in preclinical models of Alzheimer's disease - PMC](#) 12

[Study Results | ATH-1017 for Treatment of Mild to Moderate Alzheimer's Disease | ClinicalTrials.gov](#) 13

<https://clinicaltrials.gov/study/NCT04886063#study-plan> 14

[Evaluation of Metabolically Stabilized Angiotensin IV Analogs as Procognitive/Antidementia Agents - PMC](#) 15

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[https://jpet.aspetjournals.org/article/S0022-3565\(24\)25716-5/abstract](https://jpet.aspetjournals.org/article/S0022-3565(24)25716-5/abstract) 17

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[Cognitive benefits of Angiotensin IV and Angiotensin-\(1-7\): a systematic review of experimental studies - PMC](#) 21

[HGF and MET: From Brain Development to Neurological Disorders - PMC](#) 22

https://sci-hub.red/https://journals.sagepub.com/doi/10.3233/JAD-142814?url_ver=Z39.88-2003&rfr_id=ori:rid:crossref.org&rfr_dat=cr_pub%20%200pubmed 23

https://sci-hub.red/https://www.tandfonline.com/doi/10.1080/10799890701723528?url_ver=Z39.88-2003&rfr_id=ori:rid:crossref.org&rfr_dat=cr_pub%20%200pubmed 24

<https://sci-hub.red/https://www.sciencedirect.com/science/article/abs/pii/S0301008214001245?via%3Dihub> 25

https://journals.sagepub.com/doi/10.3233/JHD-231507?url_ver=Z39.88-2003&rfr_id=ori:rid:crossref.org&rfr_dat=cr_pub%20%200pubmed 26

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<https://jpet.aspetjournals.org/retrieve/pii/S0022356524467529> 30

[The Hepatocyte Growth Factor/c-Met Antagonist, Divalinal-Angiotensin IV, Blocks the Acquisition of Methamphetamine Dependent Conditioned Place Preference in Rats - PMC](#) 31

[https://jpet.aspetjournals.org/article/S0022-3565\(24\)46618-4/pdf](https://jpet.aspetjournals.org/article/S0022-3565(24)46618-4/pdf) 32

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[Autism-Associated Promoter Variant in MET Impacts Functional and Structural Brain Networks - PMC](#) 36

[Transcriptome analysis of fibroblasts from schizophrenia patients reveals differential expression of schizophrenia-related genes - PMC](#) 37

[A genetic variant that disrupts MET transcription is associated with autism - PMC](#) 38

[Conserved Subcortical and Divergent Cortical Expression of Proteins Encoded by Orthologs of the Autism Risk Gene MET - PMC](#) 39

[Safety, Tolerability, Pharmacokinetics, and Pharmacodynamics of the Positive Modulator of HGF/MET, Fosgonimeton, in Healthy Volunteers and Subjects with Alzheimer’s Disease: Randomized, Placebo-Controlled, Double-Blind, Phase I Clinical Trial - PMC](#) 40

(AI) Pooled Anecdotes

TL;DR

Across community reports, Dihexa is most often credited with measurable—usually subtle to moderate—improvements in learning, memory consolidation/recall, mental “sharpness,” and motivation/drive when used over days–weeks and sometimes stacked with other neurogenic nootropics. Typical community regimens center on low-mg intermittent dosing (examples: ~5–10 mg every other day or a few times per week). Sleep/dream changes are commonly reported but secondary to the benefit reports.

Detailed themes (aggregated anecdotal signal)

1) Typical benefit profile (what users say they get).

- Faster learning & easier consolidation. Many users describe learning new material more quickly and retaining details better across days or weeks of use.
- Improved word retrieval and fluency. Reports often mention smoother speech and quicker recall of names/terms (subtle «flow» improvements).
- Increased motivation / habit formation. Users frequently link Dihexa use with better follow-through on tasks, enhanced drive to work/study, and easier habit implementation when used alongside lifestyle changes.
- Enhanced qualitative focus (not hyperstimulation). The subjective focus is usually described as clearer and more persistent rather than jittery; many contrast it with stimulant-like effects.

2) Onset & timeline.

- Effects are commonly reported to develop over days to 2–3+ weeks of continued or scheduled use (not a single-dose, immediate magic bullet). Some users note small day-one changes; most emphasize cumulative benefit.

3) Dosing patterns people report.

- Community patterns vary but frequently cited examples: 5 mg 2–3×/week, 10 mg every other day, and 10 mg every 3–4 days when stacked. A minority report higher daily

protocols, but the modal anecdote is low, intermittent dosing. The route is usually oral; some report intranasal experimentation.

4) Stacks & synergistic reports.

- Common pairings: NSI-189, nootropic stacks (racetams etc.), and sometimes low stimulants. Many users say Dihexa feels synergistic with compounds that promote neurogenesis or synaptic plasticity—reports emphasize improved learning and longer-lasting mood/drive effects when stacked.

5) Side effects mentioned (brief).

- Sleep / vivid dreams and occasional mild sleep disruption are the most frequent negatives cited; they are often framed as manageable by dose timing. Less frequently mentioned: transient headaches or mild GI upset.

Paraphrased representative community comments

- “Running 10 mg every other day with NSI-189 — after two weeks I’m learning faster and sticking with morning habits better.”
- “Used 5 mg three times a week for months — noticed word-finding got smoother and studying felt less effortful.”
- “Not stimulant-y; it’s more like my brain’s background operating system gets quieter so tasks feel easier to execute.”
- “Stacked with NSI-189 and small caffeine doses — the combo made multi-hour deep study sessions more productive than either alone.”

Synthesis (what the anecdotal record emphasizes about benefits)

1. Primary benefit signal = learning + consolidation. The strongest recurring claim in community threads is that Dihexa helps users learn new material faster and hold it better

across days.

2. Secondary benefit = cognitive fluency and motivation. Improvements in word retrieval, smoother thought flow, and an uptick in motivation/task follow-through appear repeatedly.
3. Stacking matters. Many users report larger or more noticeable benefits when Dihexa is combined with other pro-plasticity agents (e.g., NSI-189); stacking is a common strategy in the anecdotal record.
4. Intermittent low dosing is the community norm. The majority of positive reports come from people using low, spaced dosing schedules rather than high daily dosing.